

Turning a Messy Room into a Fully Immersive VR Playground

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ABSTRACT

In this study, to enable a VR experience with an HMD even in a space with obstacles, we developed a real-time construction system of a reality-based VR space that does not impair the atmosphere of the virtual world. In addition, we aim to construct a VR space that is easier to recognize its structure by classifying “objects that are boundaries of space” and “ordinary obstacles” using a deep learning network and superimposing virtual objects corresponding to each type of real object. We implemented a real-time construction system for a VR space that reflects the distribution of objects in a real space using two depth cameras mounted on HMD, and created guidelines and applications for using the proposed system.

Index Terms: Human-centered computing—Virtual Reality—Procedural Generation; Human-centered computing—Real Walking—Locomotion—Obstacle Avoidance; Human-centered computing—Deep Learning—Segmentation

1 INTRODUCTION

In recent years, head mounted displays (HMDs) have become generally available, and anyone can easily experience virtual reality (VR). Because the user cannot know the surroundings in the real world when wearing an HMD, to use the HMD, an unobstructed space is required, and the user’s range of action is limited by the surrounding environment. By constructing a VR space according to the distribution of obstacles in the real space, a method of experiencing VR without impairing the immersive feeling even in a space with obstacles is being studied [4] [1]. In this study, we constructed a reality-based VR space in real time that does not impair the atmosphere of the virtual world even in a space with obstacles. In addition, we aim to construct a VR space that is easier to recognize by classifying “objects that are boundaries of space” and “ordinary obstacles” using a deep learning network and superimposing virtual objects corresponding to each type of real object.

2 PROPOSED METHOD

In this study, we constructed a VR space that reflects the distribution of objects in a real space in real time by arranging virtual objects in the same position as the obstacles in the real space. Fig. 1 shows the processing flow of the proposed method. The VR space is divided into voxels, and the object surface in the real space is acquired as point cloud data from the RGB-D camera mounted on the HMD. Each pixel of the acquired distance image is converted into VR spatial coordinates, the voxel to which the pixel belongs is determined, and the corresponding number of times is counted assuming that the corresponding voxel has a group of object surface points. When the count number exceeds the threshold value, it is determined that there is an obstacle in the real space at the voxel position, and VR voxel space data are updated based on virtual object superposition judgment algorithm. Based on the VR voxel space data, the virtual object is superposed, and the superposed virtual object is changed

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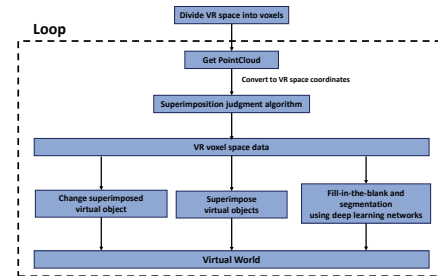


Figure 1: Proposed method

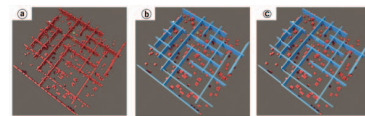


Figure 2: (a) Input, (b) output, and (c) ground truth

according to the size. In addition, the VR voxel space data are sent to a deep learned network to fill in undetermined region that cannot be measured. At the same time, segmentate on two types of “objects that are boundaries of space” and “ordinary obstacles” and the virtual objects corresponding to each type are superimposed to construct a VR space that makes spatial recognition easier to achieve.

3 COMPLEMENTING AND SEGMENTING BY DEEP LEARNING

We sent object placement information in a VR space to a deep-learned network. In addition to filling in undetermined region that could not be measured, we conducted learning with the goal of classifying objects into two types: “objects that are boundaries of space” and “ordinary obstacles”. The U-Net structure [3] is adopted as the network structure. The input is a voxel space with a size of $8 \times 128 \times 128$, and after filling in the undetermined parts, the segmented data are output in three types: “ordinary obstacle”, “wall”, and “no obstacle”. The training data set was a VR voxel space with noise added (Fig. 2a), in addition its correct label (Fig. 2c) as single set, and 31000 sets of training data, and 6400 sets of test data were used. The network was implemented using Python 3.6.10 and the deep learning library PyTorch. As an example of the learning result, the output for the input in Fig. 2a is shown in Fig. 2b. The light blue voxels indicate that it was judged to be a wall surface, and it was confirmed that the undetermined part was filled in and segmented.

4 REAL-TIME CONSTRUCTION OF VR SPACE

4.1 Device configuration

Fig. 3 shows the prototype device. For development of prototype device, an Oculus Rift S, which does not require an external tracking camera for the HMD, an Intel RealSense Depth Camera D455 (acquirable depth of 0.4 - 20 m) for distance measurements, and an Unity 2020.1.11f1 for building the VR environment were applied. RealSense cannot measure the obstacles at the user’s feet in the forward direction, and we therefore installed two units, one



Figure 3: Prototype device



Figure 4: (a)Based real space (b)Constructed VR space (with deep learning) (c)Constructed VR space (no deep learning)

in the forward direction and the other in the diagonally downward direction.

4.2 Implementation

As a result of the implementation, the VR space in Fig. 4b was constructed with respect to the real space in Fig. 4a. RealSense was set to a resolution of 848×480 and was used to measure the distance within the range of 0.4 - 7 m. The VR space was divided into voxels with a side length of 30 cm, and based on the knowledge obtained in the previous experiment [2], the virtual object was mounted 10 cm closer to the user than the actual obstacle position. In addition, the constructed VR voxel space information was sent to the trained network, and the undetermined part was filled in and segmented. The constructed VR space before filling in the holes and segmenting using the learned network is shown in Fig. 4c. As shown by the broken line in Fig. 4c, there is a large hole at a low position. Compared with Fig. 4b, it was confirmed that the learned network was used to fill in the unmeasured parts and segmented, and it was possible to construct a VR space that makes it easier to recognize the space.

5 DESIGN GUIDELINES

The following guidelines are applied for implementing content using the proposed system.

- Pre-scan : If there is no virtual object in the VR space at the start of the content, the quality of the experience may deteriorate because the user does not know how far the user can move. Therefore, the proposed system can be used to construct a VR space that reflects the object distribution in the real space in real time, however it is recommended to look around at that spot and scan the real space before the content starts (tutorial etc.).
- Limit the movement speed : Because the proposed system builds a VR space in real time, a virtual object may suddenly be superimposed in front of the user while moving. If the user moves at high speed while experiencing the content, the virtual objects might not be superimposed in time and may collide with the obstacles, and it is therefore recommended to set a penalty for high-speed movement in the content and limit the movement speed.

6 EXAMPLE EXPERIENCE

An example of implementing the application according to the design guidelines and using the proposed system is shown.

Capture Butterfly

In Capture Butterfly, the proposed system verifies that the proposal system enables a VR experience that involves movement without impairing the feeling of immersion even in a space where obstacles are present. This is a game in which the user catches butterflies

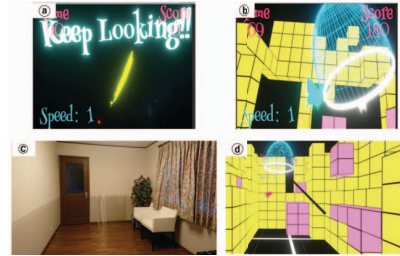


Figure 5: (a) Pre-scan, (b) Capture Butterfly, (c) real space, and (d) constructed VR space

flying around in a VR space and competes for the high scores. After the tutorial phase, which describes the contents of the game, the game has the main task of catching butterflies. As a pre-scan, during the tutorial phase, it is necessary to continue looking the butterflies that sway up and down and go around the player for 30 s (Fig. 5a). The space is scanned while watching the butterflies, and the VR space is constructed based on the scanned data at the start of the game. During the 90 s playing phase, the player attempts catch three types of butterflies with different points depending on the color and competes for the final score (Fig. 5b). Because the target butterfly disappears when the player moves above a certain speed, the player needs to move slowly, and the movement speed is therefore limited. When playing in the real space of Fig. 5c, the VR space of Fig. 5d was constructed, and the player could enjoy the game while walking even in a space with obstacles.

7 CONCLUSION AND FUTURE WORK

In this study, to enable a VR experience with an HMD even in a space with obstacles, we proposed a method for constructing a reality-based VR space in real time that does not impair the world view even in a space with obstacles.

Based on the point cloud data acquired using the RGB-D camera, we propose and implement a system that installs a virtual object at the same position as an obstacle in the real space, and fills in any holes while applying segmentation from a deep learning network. We aimed to build a VR space that makes space recognition easier to achieve.

As a result of the implementation, we were able to construct a real-space-based VR space in real time that does not impair the atmosphere of the virtual world even in a location with obstacles. We created content using the proposed system and confirmed that it is possible to enjoy content that involves walking even in obstacle-laden spaces.

Finally, the proposed system cannot properly present moving object such as human. We also plan to improve system so that user can experience VR in space with people.

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